Underhung Springs.

BY J. P. KELLEY.

The method of hanging the driving springs and the equalizers of any locomotive is always of interest to the engineer, because he is, as a rule, desirous of knowing the extent of the effect which the failure of a spring or hanger will have in causing delays, when such failures occur on the road.

Nearly every engineer knows that the object of hanging the weight of the boiler and its attachments on springs is to permit the engine to be run smoothly at almost any rate of speed of which it is capable, over rough and uneven spots in the track, without severe jar or shock.

When it was the rule to employ much lighter engines than we have nowadays it was customary to place the driving springs over the driving boxes, allowing them to rest on spring stirrups, that in turn rested upon the top of the driving box, and then by means of hangers and equalizers to suspend the weight of the boiler from them.

As the demand for heavier engines and more boiler capacity increased, the space above the driving boxes, which had been used for the springs and hangers, had to be given up to make room for the larger boiler, which now generally spreads out over the frame at the firebox end, and a location under the driving boxes is now, in many modern designs of locomotives, used for placing the driving springs, hangers and the equalizers.

On eight-wheel engines, of which we will treat only, the arrangement of the springs and equalizers when underhung is not radically different from what we find it when they are overhung, as may be seen by reference to Figs. 1 and 2 accompanying this article, except that it requires an entirely different scheme from that of the ordinary spring stirrups, or saddle, for holding the driving springs in place and for transferring their load to the driving boxes.

Fig. 1 shows the form of hanger, or we might term it underhung spring saddle, that is employed on some engines. It is attached to the driving box, extends downward and encircles the spring at the band. This form of spring saddle prevents the removal of the driving box cellar when it is necessary to pack the driving box, unless the engine is jacked up; otherwise it has proved very satisfactory.

In Fig. 2 the springs are underhung, but the weight of the engine is all transferred to the top of the driving box on account of the long hangers that are connected to the equalizer beams that rest on the top of the driving box and from which the weight is suspended.

A little study of the figures will make clear the manner in which the weight is distributed to, and equalized on, the driving boxes. There are many other methods of hanging the springs under the driving boxes and different designs for the frame hanger brackets, but in principle they are all similar to those shown in the figures.

One thing may be said of the underhung spring, and that is, if it breaks, or if any of its hangers break, there is not the likelihood of the end of the spring getting far out of place or of the equalizer flying up so far that it will be a difficult job to get it back where it belongs. The most difficult part of fixing up a broken spring or hanger on the road, on engines that have underhung springs, so as to get in without heating the driving boxes, as I look at it, is jacking the engine up high enough to get the required amount of blocking in between the top of the driving box and the frame, when necessary to hold the engine level, or in between the frame and the end of the driving spring, or frame and the end of the equalizer, as the case may require.

A few words here as to how to raise the engine up when it is necessary to do so, will not be out of place.

Time is, next to safety, the most important consideration on a railroad as elsewhere; therefore before doing a whole lot of unnecessary work, look the situation over carefully to determine just how much is needed to be done. It may so happen that if a hanger breaks, or a spring, if your terminal is not far away, you can get along without jacking up or blocking up, and so need not cause any serious delay to your train. But if you find that it is necessary to jack up, then I believe in using a wedge to raise the engine with; a screw-jack—well, screw-jacks nowadays are seldom in good enough condition to work the screw in and out without any load on them, not to speak of raising a 50-ton locomotive with them.

Let us suppose the hanger 1 breaks, in either plan, as shown in Figs. 1 and 2. This will let the forward end of the driving spring move up in the hanger bracket

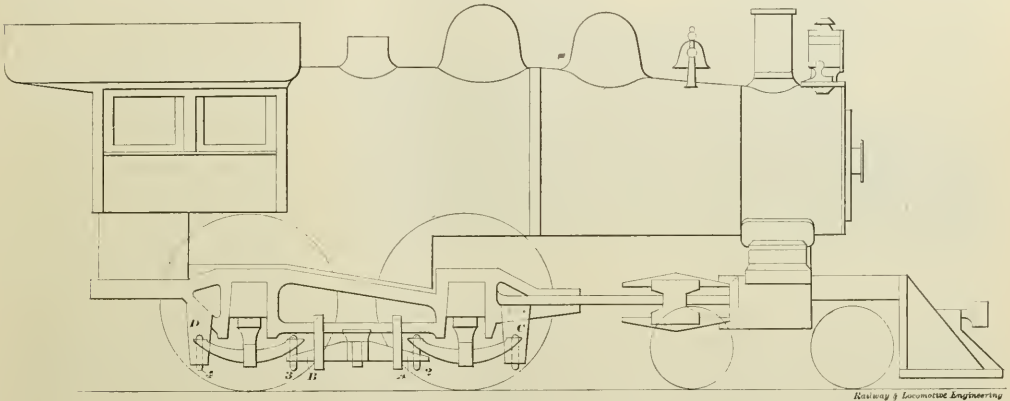
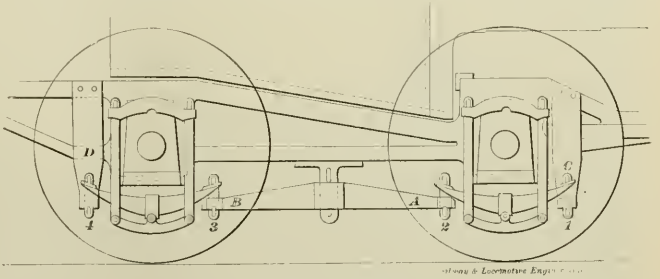


Fig. 1



UNDERHUNG SPRINGS.

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Editorial Change.

In the beginning of June Mr. George S. Hodgins became a member of the editorial staff of RAILWAY AND LOCOMOTIVE ENGINEERING. We bespeak for him the favorable consideration of our numerous friends. Mr. Hodgins is not a stranger to the readers of this journal, as he was for years a correspondent whose writings were read with interest. He received a sound engineering education in the shop and in the drawing office and held several important positions on one of our leading railways. Mr. Hodgins was for some time editor of the *Railroad Digest*, so he is not new to the routine work of the office of a railway journal.

Valve Oil.

In one pint of valve oil fed through a lubricator in good repair at the rate of 5 drops feed per minute, there is not less than 6,600 drops. The valve oil on leaving the feed nozzle of the lubricator passes drop by drop through the water in the sight feed glasses and in that condition, drop by drop reaches the steam chest. The valve oil on passing into the oil pipe is not sprayed or atomized as is supposed by many, but travels entirely by gravity, and it will therefore appear plain that anything which may interfere with or delay its passage will cause the valves and cylinders to become dry. The oil pipes leading from the lubricator to the steam chests, or air pump, should have a gradual fall from the lubricator to these parts. When the oil is fed at the rate of 5 drops feed per minute, it will require 12 minutes for the oil to flow from the lubricator to the steam chest on a consolidation engine, and it is a good practice before putting an engine in service, to guard against possible defects or pockets in the oil pipes, by starting the lubricator to work at 5 drops feed per minute, disconnecting the oil pipes at the steam chests, noting the length of time it requires for the oil to flow from the lubricator to this point. If the time required is in excess of 15 minutes it is safe to conclude that there is a pocket in the oil pipe, which, of course, ought to be remedied, and as we know that it requires this time for the oil, under the very best of conditions, to flow from the lubricator to the steam chest, engineers should be instructed on this point, so that they may start the lubricator working a sufficient length of time before starting their trains as will allow of the oil reaching the steam chest about the time or a little before the engine starts on its trip. Knowing as we do that the oil will not flow up hill, it would seem that the best results in air pump lubrication might be obtained by attaching the lubricator pipe to the highest point on the steam end of the air pump. When attached to the pump at that point, 1 drop feed of oil per minute is all that is required to lubricate a 9½-inch or a 11-inch pump.

Five drops feed per minute will ordinarily be found sufficient for the largest engine and heaviest service; for smaller engines or light service a proportionately slower feed of oil will be found sufficient. Intelligent and economical results cannot be obtained unless enginemen know the rate at which the oil is being fed to the air pump and steam chests. Before commencing to ascend a heavy grade, it is a good idea to quicken the feed of oil a few minutes before reaching the foot of the grade, and as it has been conclusively proved that when a full throttle is being used on high pressure engines, especially at a low rate of speed, that the oil does not flow continuously to the steam chest. It is an excellent practice to occasionally close the throttle partially to allow such oil as has been held in suspension in the oil pipes to flow into the steam chest, the object of this being to momentarily reduce the pressure in the steam chest below that in the oil pipes. If a small quantity of oil is put in the lubricator, time should be allowed for the condensation to fill the cup and raise the oil to the top of the feed pipes or a sufficient quantity of water in addition to the oil applied should be added before the feed valves are opened. Immediately after the lubricator is filled the steam or water valve should be opened; the steam valve should always be opened first and opened full and should be shut last. If this practice is followed the water in the feed glasses will remain clear and there will be no loss of oil by syphoning. If the feed glasses become dirty they may be cleaned by rubbing a little glycerine on the inside of the glasses. If the lubricator is in good repair there will be very little variation in its feed between using steam and with the throttle closed; if the feed of the lubricator slackens when the throttle is opened or quickens when the throttle is closed, it may be concluded that the choke plugs have become enlarged and in that case satisfactory results cannot be obtained until the defects have been remedied. In most of the lubricators now on the market the choke plugs are located in the upper feed arms, and by disconnecting the oil pipe at that point can be conveniently reached and easily removed with a small monkey wrench, and as they are sold for about 50 cents a dozen, the expense attached to their renewal is very slight. At times it will be noticed that the drop of oil forming at the outlet of the feed nozzle takes a balloon-like shape and is slow in separating itself from the nozzle. This is due to a slight corrosion occurring at the tip and inside of the nozzle, and may be easily removed with a pin or any other small device. If the opening in the choke plug becomes stopped up, it can be opened by shutting the steam valve of the lubricator and opening the throttle valve of the engine, the pressure passing out through the oil pipe will force the substance in the lubricator down into the feed glass. If the feed valves become stopped up they can be opened by closing the condenser or water valve and opening the drain valve; the pressure will then force everything up into the lubricator.

In order to prevent small particles of waste or other matter getting into the oil cans in use on the locomotive, strainers should be used at all oil houses in the funnels used at those places.

The temperature of steam at 200 pounds pressure is 380 degrees; the fire test of valve oil is 550 degrees. The object of having the fire test of valve oil so high is to insure the retention of its lubricating qualities, even when subjected to a very high temperature. When valve oil is used extravagantly it is thrown out of the cylinders, lodged in the exhaust pipes, where it soon forms a very thick coating, reducing the size of the exhaust pipes and nozzles, increases the consumption of coal and interferes with the free working of the engine.

By the fire test of an oil is meant the temperature at which that oil will ignite. The fire test of valve oil is 550 degrees, the fire test of winter engine oil about 300 degrees, the fire test of summer engine oil about 400 degrees, the fire test of winter car oil is about 300 degrees, the fire test of summer car oil about 400 degrees, the fire test of winter coach oil about 300 degrees, the fire test of summer coach oil about 400 degrees, the fire test of signal